

Hardik Abrol

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EDUCATION

Thapar Institute of Engineering and Technology
Bachelor of Engineering, Computer Engineering | CGPA: 7.62/10

Patiala, India
Aug 2023 – May 2027

Kartar Public School, Barnoti
High School (PCM) | 94.6%

Kathua, Jammu and Kashmir
Apr 2021 – Apr 2023

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, TypeScript, SQL
Machine Learning & AI: PyTorch, Scikit-learn, CatBoost, HuggingFace, LangChain, SHAP
Web & Backend: FastAPI, Flask, SQLAlchemy, React, Next.js
Data & Databases: Pandas, NumPy, MySQL, PL/SQL
Developer Tools: Git, Streamlit

PROJECTS

Athlyt – AI Fitness Coach — *Next.js, TypeScript, FastAPI, SQLAlchemy* — [GitHub] [Demo]
Built a production-quality full-stack fitness coaching platform with personalized workout plans, nutrition tracking, progress analytics, and workout session management.

- Built a **full-stack application** with a Next.js 15/TypeScript/Tailwind CSS frontend and a **FastAPI + SQLAlchemy 2.0** backend, designed with a Postgres-portable schema.
- Implemented secure **JWT-based authentication** and onboarding, computing BMI and daily calorie estimates from user profile data.
- Designed a **modular workout recommendation engine** that maps user profiles to training splits (PPL, Upper/Lower, Full Body, and more) via a rule pipeline built to be swapped for an ML model without touching other code.
- Established **CI/CD pipelines** for both frontend and backend backed by a **220-test suite**, and deployed the app live on Vercel and Render.

Mental Health Predictor App — *Python, CatBoost, Scikit-learn, Streamlit, SHAP* — [GitHub] [Demo]
Implemented an interpretable classification system to identify individuals needing clinical mental health support.

- Developed an **interpretable ML classification system** using CatBoost to predict the likelihood of requiring mental health treatment based on high-dimensional survey features.
- Engineered a preprocessing pipeline including **Cramér's V correlation analysis** for feature selection and Chi-square tests to ensure statistical significance of categorical variables.
- Integrated **model explainability** using SHAP values to surface top contributing factors for each prediction, increasing clinical trust and transparency.
- Deployed a responsive **Streamlit web application** featuring interactive data dashboards and real-time inference, successfully serving live predictions to over **200+ test users**.

EXPERIENCE

GS Tech Technology Pvt Ltd (AdSpyder)
Intern - Web Developer (Remote)

Jun 2026 – Present
Bengaluru, Karnataka

- Engineered responsive web applications and optimized dynamic user interfaces to improve client-side navigation and core system responsiveness.
- Collaborated asynchronously with cross-functional development teams to deploy scalable features, integrating frontend layouts with backend APIs and internal web services.
- Refactored legacy codebases and troubleshooted technical issues, reducing page load latencies and ensuring data integrity across web platforms.

CERTIFICATIONS

NVIDIA: Fundamentals of Deep Learning
Whizlabs — Coursera [Credential]

Feb 2026
Course Certificate

- Built and trained deep learning models from scratch, implementing CNNs for computer vision tasks, model optimization, data augmentation, and transfer learning.

Generative AI with Large Language Models Specialization
DeepLearning.AI & AWS — Coursera [Credential]

Apr 2026
3-Course Series

- Gained hands-on experience with the typical GenAI lifecycle, including scoping, selecting, and fine-tuning Large Language Models (LLMs) for specific use cases.